

HW6_TestThat

AUTHOR

Rebeca Adam, Natasha Atkins, Elijah Khan

Description

This document demonstrates how to use the Assignment 5 functions to create sample neighborhood data and calculate flood risk, social vulnerability, insurance retreat risk, and cumulative risk.

```
## Load Functions  
library(here)
```

here() starts at /Users/rebecaadam/Desktop/R Projects/HW6_TestThat

```
source(here("Create Neighborhood Data.R"))  
source(here("compute_flood_risk.R"))  
source(here("calculate_vulnerability.R"))  
source(here("calculate_Insurance_Retreat.R"))  
source(here("cumulative_risk_function.R"))
```



Example Neighborhood Data

```
neighborhood_data <- create_neighborhood_data(n = 10)  
  
head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	844.1719	840.6888	72.55186	168.28172	0.6194279	0.45961869
2	b	2559.1842	929.8173	73.14784	109.43566	0.9811162	0.58984995
3	c	1483.6802	675.1625	246.43853	68.27611	0.8508357	0.09106137
4	d	652.5330	618.1785	20.74647	78.19544	0.9956650	0.09675505
5	e	585.1478	704.4767	131.00122	90.32903	0.9310591	0.47441980

```

6   f  605.1514      787.8410 249.94959      178.77841 0.2155937 0.14202647
   no_insurance  renters  elderly  children  disabled percent_nonrenewal
1   0.4099309 0.49457482 0.07473866 0.9366198 0.5022319      0.92277533
2   0.3173319 0.81176017 0.11458981 0.3000848 0.8647623      0.71471549
3   0.3445494 0.43057306 0.63172370 0.1555766 0.3010923      0.87221380
4   0.7501504 0.23211049 0.40227259 0.4562626 0.1001559      0.78636973
5   0.0670817 0.27419875 0.66864689 0.8255593 0.3121529      0.09694907
6   0.9129826 0.02152931 0.06036751 0.8520760 0.1716858      0.82912711
   premium_increase past_claims
1   0.974348444      43.87218
2   0.192657972      171.96572
3   0.001901226      176.15612
4   0.209459605      156.34585
5   0.157664691      117.11528
6   0.137651914      174.00118

```

Calculating the Flood Risk

```

neighborhood_data$flood_risk <- compute_flood_risk(
  elevation = neighborhood_data$elevation,
  distance_coast = neighborhood_data$distance_coast,
  mean_rain = neighborhood_data$mean_rain,
  num_past_floods = neighborhood_data$num_past_floods
)

```

```
head(neighborhood_data)
```

```

   city elevation distance_coast mean_rain num_past_floods  low_inc  minority
1   a  844.1719      840.6888  72.55186      168.28172 0.6194279 0.45961869
2   b 2559.1842      929.8173  73.14784      109.43566 0.9811162 0.58984995
3   c 1483.6802      675.1625 246.43853      68.27611 0.8508357 0.09106137
4   d  652.5330      618.1785  20.74647      78.19544 0.9956650 0.09675505
5   e  585.1478      704.4767 131.00122      90.32903 0.9310591 0.47441980
6   f  605.1514      787.8410 249.94959      178.77841 0.2155937 0.14202647
   no_insurance  renters  elderly  children  disabled percent_nonrenewal
1   0.4099309 0.49457482 0.07473866 0.9366198 0.5022319      0.92277533

```

2	0.3173319	0.81176017	0.11458981	0.3000848	0.8647623	0.71471549
3	0.3445494	0.43057306	0.63172370	0.1555766	0.3010923	0.87221380
4	0.7501504	0.23211049	0.40227259	0.4562626	0.1001559	0.78636973
5	0.0670817	0.27419875	0.66864689	0.8255593	0.3121529	0.09694907
6	0.9129826	0.02152931	0.06036751	0.8520760	0.1716858	0.82912711

	premium_increase	past_claims	flood_risk
1	0.974348444	43.87218	0.5045857
2	0.192657972	171.96572	0.3033852
3	0.001901226	176.15612	0.4488398
4	0.209459605	156.34585	0.3630172
5	0.157664691	117.11528	0.4371443
6	0.137651914	174.00118	0.6676659

Calculating the Social Vulnerability

```
neighborhood_data$vulnerability_score <- calculate_vulnerability(
  low_inc = neighborhood_data$low_inc,
  minority = neighborhood_data$minority,
  no_insurance = neighborhood_data$no_insurance,
  renters = neighborhood_data$renters,
  elderly = neighborhood_data$elderly,
  children = neighborhood_data$children,
  disabled = neighborhood_data$disabled
)
```

Loading required package: usethis

Attaching package: 'devtools'

The following object is masked from 'package:testthat':

test_file

```
head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	844.1719	840.6888	72.55186	168.28172	0.6194279	0.45961869
2	b	2559.1842	929.8173	73.14784	109.43566	0.9811162	0.58984995
3	c	1483.6802	675.1625	246.43853	68.27611	0.8508357	0.09106137
4	d	652.5330	618.1785	20.74647	78.19544	0.9956650	0.09675505
5	e	585.1478	704.4767	131.00122	90.32903	0.9310591	0.47441980
6	f	605.1514	787.8410	249.94959	178.77841	0.2155937	0.14202647

	no_insurance	renters	elderly	children	disabled	percent_nonrenewal
1	0.4099309	0.49457482	0.07473866	0.9366198	0.5022319	0.92277533
2	0.3173319	0.81176017	0.11458981	0.3000848	0.8647623	0.71471549
3	0.3445494	0.43057306	0.63172370	0.1555766	0.3010923	0.87221380
4	0.7501504	0.23211049	0.40227259	0.4562626	0.1001559	0.78636973
5	0.0670817	0.27419875	0.66864689	0.8255593	0.3121529	0.09694907
6	0.9129826	0.02152931	0.06036751	0.8520760	0.1716858	0.82912711

	premium_increase	past_claims	flood_risk	vulnerability_score
1	0.974348444	43.87218	0.5045857	0.4995918
2	0.192657972	171.96572	0.3033852	0.5684993
3	0.001901226	176.15612	0.4488398	0.4007732
4	0.209459605	156.34585	0.3630172	0.4333389
5	0.157664691	117.11528	0.4371443	0.5075883
6	0.137651914	174.00118	0.6676659	0.3394659

Calculating Insurance Retreat Score

```
neighborhood_data <- calculate_insurance_retreat_score(neighborhood_data)
```

```
head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	844.1719	840.6888	72.55186	168.28172	0.6194279	0.45961869
2	b	2559.1842	929.8173	73.14784	109.43566	0.9811162	0.58984995
3	c	1483.6802	675.1625	246.43853	68.27611	0.8508357	0.09106137
4	d	652.5330	618.1785	20.74647	78.19544	0.9956650	0.09675505
5	e	585.1478	704.4767	131.00122	90.32903	0.9310591	0.47441980
6	f	605.1514	787.8410	249.94959	178.77841	0.2155937	0.14202647

	no_insurance	renters	elderly	children	disabled	percent_nonrenewal
--	--------------	---------	---------	----------	----------	--------------------

1	0.4099309	0.49457482	0.07473866	0.9366198	0.5022319	0.92277533
2	0.3173319	0.81176017	0.11458981	0.3000848	0.8647623	0.71471549
3	0.3445494	0.43057306	0.63172370	0.1555766	0.3010923	0.87221380
4	0.7501504	0.23211049	0.40227259	0.4562626	0.1001559	0.78636973
5	0.0670817	0.27419875	0.66864689	0.8255593	0.3121529	0.09694907
6	0.9129826	0.02152931	0.06036751	0.8520760	0.1716858	0.82912711

	premium_increase	past_claims	flood_risk	vulnerability_score
1	0.974348444	43.87218	0.5045857	0.4995918
2	0.192657972	171.96572	0.3033852	0.5684993
3	0.001901226	176.15612	0.4488398	0.4007732
4	0.209459605	156.34585	0.3630172	0.4333389
5	0.157664691	117.11528	0.4371443	0.5075883
6	0.137651914	174.00118	0.6676659	0.3394659

	insurance_retreat_score
1	0.8027217
2	0.5349151
3	0.5258021
4	0.5546776
5	0.2189608
6	0.5607128

Calculate Cumulative Risk

```
neighborhood_data$cumulative_risk_index <- compute_risk_index(
  flood_vulnerability = neighborhood_data$flood_risk,
  social_vulnerability = neighborhood_data$vulnerability_score,
  non_coverage_risk = neighborhood_data$insurance_retreat_score
)

head(neighborhood_data)
```

	city	elevation	distance_coast	mean_rain	num_past_floods	low_inc	minority
1	a	844.1719	840.6888	72.55186	168.28172	0.6194279	0.45961869
2	b	2559.1842	929.8173	73.14784	109.43566	0.9811162	0.58984995
3	c	1483.6802	675.1625	246.43853	68.27611	0.8508357	0.09106137
4	d	652.5330	618.1785	20.74647	78.19544	0.9956650	0.09675505

5	e	585.1478	704.4767	131.00122	90.32903	0.9310591	0.47441980
6	f	605.1514	787.8410	249.94959	178.77841	0.2155937	0.14202647
		no_insurance	renters	elderly	children	disabled	percent_nonrenewal
1		0.4099309	0.49457482	0.07473866	0.9366198	0.5022319	0.92277533
2		0.3173319	0.81176017	0.11458981	0.3000848	0.8647623	0.71471549
3		0.3445494	0.43057306	0.63172370	0.1555766	0.3010923	0.87221380
4		0.7501504	0.23211049	0.40227259	0.4562626	0.1001559	0.78636973
5		0.0670817	0.27419875	0.66864689	0.8255593	0.3121529	0.09694907
6		0.9129826	0.02152931	0.06036751	0.8520760	0.1716858	0.82912711
		premium_increase	past_claims	flood_risk	vulnerability_score		
1		0.974348444	43.87218	0.5045857		0.4995918	
2		0.192657972	171.96572	0.3033852		0.5684993	
3		0.001901226	176.15612	0.4488398		0.4007732	
4		0.209459605	156.34585	0.3630172		0.4333389	
5		0.157664691	117.11528	0.4371443		0.5075883	
6		0.137651914	174.00118	0.6676659		0.3394659	
		insurance_retreat_score	cumulative_risk_index				
1		0.8027217		0.5526104			
2		0.5349151		0.4303449			
3		0.5258021		0.4456447			
4		0.5546776		0.4184012			
5		0.2189608		0.4242617			
6		0.5607128		0.5404404			

The Final Output

```
final_output <- neighborhood_data[
  c("city", "flood_risk", "vulnerability_score",
    "insurance_retreat_score", "cumulative_risk_index")
]
```

```
final_output
```

	city	flood_risk	vulnerability_score	insurance_retreat_score
1	a	0.5045857	0.4995918	0.8027217
2	b	0.3033852	0.5684993	0.5349151

3	c	0.4488398	0.4007732	0.5258021
4	d	0.3630172	0.4333389	0.5546776
5	e	0.4371443	0.5075883	0.2189608
6	f	0.6676659	0.3394659	0.5607128
7	g	0.4199511	0.5239159	0.5033387
8	h	0.7705336	0.4544770	0.8250255
9	i	0.4798994	0.5488835	0.3973747
10	j	0.8966436	0.6134117	0.5116291

	cumulative_risk_index
1	0.5526104
2	0.4303449
3	0.4456447
4	0.4184012
5	0.4242617
6	0.5404404
7	0.4685040
8	0.6742634
9	0.4891400
10	0.7380639

Explanation

This example shows how all five of the Assignment 5 modules can be used together to assess flood/social vulnerability at the neighborhood level. First, sample neighborhood data is generated using `create_neighborhood_data()`. Then, flood risk, social vulnerability, and insurance retreat score are calculated separately, but require inputs from the following modules to run accordingly. Finally, each metric is combined into a cumulative risk index through the `compute_risk_index()` function. This provides a measure of the overall neighborhood risk!